Appendix I. Debt Sustainability Analysis-Update[5]

October 15, 2010

This updated debt sustainability analysis (DSA) yields results similar to those reported in the June 2010 DSA—the risk of debt distress continues to be regarded as low. The analysis continues to be based on the assumptions that macroeconomic policies will be strengthened to stabilize the external current account deficit at sustainable levels and that a prudent borrowing strategy will be maintained. Under those assumptions Honduras' debt would remain low and sustainable, with all debt indicators remaining below their indicative thresholds and robust under various stress tests. However, debt dynamics would weaken if policies deviate from the program and/or a permanent adverse shock to GDP growth.

KEY ASSUMPTIONS

Strengthened macroeconomic policies. As in the 2010 Article IV consultation documents, the scenario continues assuming that macroeconomic policies are strengthened starting in the second half of 2010, and remain strong over the medium term. In addition to a tax reform approved last April that would yield revenue equivalent to 1½–2½ percent of GDP (on an annual basis) over three years, the baseline also assumes the adoption of additional measures, which the authorities are currently considering. These include controlling the wage bill, reduce subsidies, and adjust electricity tariffs to reduce the public sector deficit to about 2 percent of GDP over the medium term.

Real GDP will grow by 4 percent per year helped by improved performance in exports of commodities, in the *maquila*[6] sector, and remittances. Also, growth prospects are assumed to be enhanced by higher public investment, which is assumed to reach 6.5 percent of GDP over the medium term (compared to 5.3 percent in 2007–09).

The overall fiscal deficit of the public sector will stabilize at about 2 percent of GDP.[7] Public sector revenues and grants are projected to increase to around 25 percent of GDP by 2012. The composition of expenditure would be tilted toward investment and poverty reduction, and after 2013 would increase broadly in line with nominal GDP. The fiscal deficit would be financed mostly by concessional resources from multilateral and bilateral creditors.

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[5] The DSA update has been prepared jointly by the IMF and World Bank staff.

[6] A *maquila* or *maquiladora* is a manufacturing plant that imports and assembles duty-free components for export. Maquila exports represent 50 percent of total exports and are mainly related to textiles and confections.

[7] The consolidated public sector includes the central government, local governments, decentralized agencies, social security institutions, the central bank, and public enterprises.

The noninterest external current account deficit is projected to decline below 6 percent of GDP in the long term. This would be the result of favorable dynamics in the trade balance, tourism, and remittances. The current account deficit would be largely financed by official sources and foreign direct investment (FDI), and support the projected expansion in *maquila* and tourism.

Annual export growth would reach nearly 8 percent over the long run. A projected decline in the prices of Honduras' main commodities exports (coffee and bananas) is expected to be offset by continued productivity gains and higher access to international markets (through CAFTA and a trade agreement with the European Union, which would become effective in 2012).

Annual import growth would stabilize at about 7 percent per year. Nonfuel imports are projected to increase moderately, in line with domestic demand growth and *maquila* activity. Fuel imports volumes are assumed to grow in line with real GDP.

Non-concessional financing. The program for October 2010-March 2012 envisages a continuous ceiling for contracting non-concessional financing of US$350 million;[8] of the total, US$280 million corresponds to a new multi-sector loan from the Central American Bank for Economic Integration (CABEI).[9] Also, as in the 2010 Article IV, it is assumed that Honduras would progressively tap additional non-concessional sources of financing (which would rise from 5 percent to 23 percent of total disbursements in the long-run), mostly loans from bilateral (Paris Club and non-Paris Club countries) and commercial sources.

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[8] Honduras is classified as a low-capacity, low-vulnerability country, and is allowed to have untied non-concessional financing.

[9] Lending from CABEI is on non-concessional terms. However, recently CABEI lending to Honduras for some projects (e.g. infrastructure) was combined with concessional lending from the other institutions (e.g. World Bank and IADB) in order to reach of concessionality of 35 percent. For the purpose of this DSA update, however, the US$280 million lending has been considered non-concessional.

Fund arrangements. The Honduran authorities are requesting Fund support in the form of 18-month SBA/SCF blended arrangements in the amount of SDR129.5 million (100 percent of quota). An initial disbursement (combined SBA/SCF) of SDR35.61 million (27.5 percent of quota) would become available upon Board approval. Even though the authorities plan to treat the arrangements as precautionary, it will be assumed for this DSA update that the country uses all Fund resources as programmed.

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

Honduras' external debt would remain sustainable. All external debt indicators remain below their thresholds[10], with the PV of external debt projected to stay at below 22 percent of GDP over the entire horizon of the analysis (Table 3a and Figure 1).

External debt indicators remain below their indicative, policy-related thresholds under all stress tests and alternatives scenarios (Figure 1). Under the historical scenario, debt burden indicators are lower than those under the baseline scenario because the projected averages (from 2011) of FDI and real GDP growth are higher than the estimated values for 2010. On the other hand, the noninterest current account average is lower than the one for 2010.

PUBLIC SECTOR DEBT SUSTAINABILITY ANALYSIS

Honduras' public sector debt would remain low and sustainable. Public sector debt would remain at about 25 percent of GDP (21 percent in PV terms). As the fiscal deficit would progressively decline to about 2 percent of GDP and growth rebounds to its long-term rate, the sustainable primary deficit would be about 1 percent of GDP.

Stress Tests

The alternative scenarios and stress tests show that public sector debt dynamics are vulnerable to adverse shocks (exogenous or policy-related). The status quo stress test, which assumes a primary fiscal deficit at the 2011 level in terms of GDP until the end of the projection period, as well as a permanent decline in real GDP growth, would place the PV of public debt-to-GDP ratio on an upward trend (Table 2a). Also, a one-time real depreciation of 30 percent (e.g. under a severe balance of payments correction) or a 10 percent of GDP increase in other debt-creating flows (e.g. in case of severe financial sector distress that leads to the realization of contingent liabilities) would rise the PV of public debt-to-revenue ratio over 100 percent in the long term, compared to 85 percent under the baseline scenario.

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[10] Based on the recent update, the three-year average of the Honduras' CPIA index (between 2007 and 2009) is 3.69. In this context, the debt burden thresholds correspond to the ones associated with a medium policy performance (a rating above 3.75 corresponds to strong performance). The latter represents a deterioration compared to the previous average (2006-2008), which was 3.8 and was associated with a strong policy performance.

CONCLUSION

Based on the above analysis, Honduras' debt is assessed to be at low risk of debt distress. Assuming consistent pursuit of strengthened macroeconomic policies, Honduras' debt indicators would remain below their indicative thresholds and broadly resilient to adverse shocks. Under alternative policies, however, public sector debt dynamics would weaken. Honduras' favorable debt outlook reflects the low initial levels of public debt, which resulted in large debt relief granted to Honduras during 2000s.

Given that Honduras would progressively tap more nonconcessional sources of financing, it will be important to build on the existing capacity to manage debt and to formulate a strategy linked to medium-term fiscal framework. Authorities are committed to strengthen their debt management capacity, including with additional new technical assistance.

Table 1a.Honduras: Public Sector Debt Sustainability Framework, 2007-2030
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2007	2008	2009	Average	Standard Deviation	2010	2011	2012	2013	2014	2015	2010-15 Average	2020	2030	2016-30 Average
Public sector debt 1/	19.7	20.6	23.7			27.7	29.4	29.9	30.4	30.4	30.0		27.1	25.0	
o/w foreign-currency denominated	16.4	17.1	17.2			19.2	21.3	22.5	23.1	23.1	22.8		20.5	17.7	0.71
Change in public sector debt	-12.0	0.9	3.1			4.0	1.7	0.5	0.5	0.0	-0.4		-0.4	-0.1	
Identified debt-creating flows	-3.0	-1.4	4.3			2.1	1.6	0.6	0.4	-0.1	-0.2		-0.1	-0.2	
Primary deficit	0.1	0.2	4.0	0.7	2.4	2.6	1.7	0.6	0.5	0.5	0.5	1.1	0.7	0.7	0.6
Revenue and grants	24.4	26.3	24.8			24.5	24.5	25.0	25.0	25.0	25.0		25.0	25.1	
of which: grants	1.6	2.1	1.9			1.7	1.0	1.5	1.5	1.5	1.5		0.6	0.7	
Primary (noninterest) expenditure	24.5	26.5	28.9			27.2	26.2	25.6	25.6	25.5	25.6		25.7	25.8	
Automatic debt dynamics	-3.0	-1.5	0.2			-0.6	-0.1	0.0	-0.2	-0.6	-0.7		-0.7	-0.8	
Contribution from interest rate/growth differential	-2.0	-0.6	0.8			0.1	-0.2	-0.5	-0.6	-0.6	-0.7		-0.7	-0.8	
of which: contribution from average real interest rate	-0.1	0.1	0.4			0.7	0.7	0.6	0.6	0.6	0.5		0.4	0.2	
of which: contribution from real GDP growth	-1.9	-0.8	0.4			-0.6	-0.9	-1.1	-1.2	-1.2	-1.2		-1.1	-1.0	
Contribution from real exchange rate depreciation	-1.0	-0.9	-0.5			-0.7	0.1	0.5	0.4	0.0	-0.1		...	...	
Other identified debt-creating flows	-0.1	-0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.1	-0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-9.1	2.3	-1.2			1.9	0.1	0.0	0.1	0.1	-0.2		-0.4	0.1	
Other Sustainability Indicators															
PV of public sector debt	19.1	17.8	18.9			22.4	24.0	24.6	25.1	25.2	25.0		22.8	21.4	
o/w foreign-currency denominated	15.9	14.3	12.4			13.9	16.0	17.2	17.8	17.9	17.7		16.1	14.1	
o/w external	15.9	14.3	12.4			13.9	16.0	17.2	17.8	17.9	17.7		16.1	14.1	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	2.8	1.6	7.9			5.9	6.0	4.8	4.6	4.7	4.8		4.8	4.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	78.4	67.6	76.0			91.1	98.2	98.4	100.4	100.5	99.7		90.9	85.0	
PV of public sector debt-to-revenue ratio (in percent)	83.8	73.4	82.1			97.8	102.4	104.7	106.9	106.9	106.0		93.3	87.5	
o/w external 3/	69.5	59.0	54.0			60.7	68.0	73.0	75.8	76.2	75.4		66.1	57.6	
Debt service-to-revenue and grants ratio (in percent) 4/	11.1	5.4	13.9			13.5	17.5	16.8	16.4	16.9	17.2		16.7	15.9	
Debt service-to-revenue ratio (in percent) 4/	11.9	5.8	15.0			14.5	18.3	17.9	17.5	18.0	18.3		17.1	16.3	
Primary deficit that stabilizes the debt-to-GDP ratio	12.1	-0.7	1.0			-1.4	0.0	0.1	0.1	0.5	0.9		1.1	0.8	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.2	4.0	-1.9	3.8	2.9	2.4	3.5	4.0	4.0	4.0	4.0	3.7	4.0	4.0	4.0
Average nominal interest rate on forex debt (in percent)	1.5	2.2	1.9	1.8	0.4	2.3	1.7	1.8	1.8	2.0	1.9	1.9	2.0	2.1	2.0
Average real interest rate on domestic debt (in percent)	5.8	3.8	6.6	7.6	6.9	6.9	8.2	8.2	8.5	8.2	7.2	7.9	5.0	2.0	4.2
Real exchange rate depreciation (in percent, + indicates depreciation)	-3.7	-5.8	-3.1	-0.6	8.3	-4.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	6.9	8.5	4.4	6.5	9.5	4.9	4.0	5.6	5.1	5.2	5.5	5.1	5.3	5.3	5.3
Growth of real primary spending (deflated by GDP deflator, in percent)	11.6	12.6	6.7	8.9	13.7	-3.6	-0.1	1.7	3.7	3.8	4.2	1.6	4.2	4.1	4.1
Grant element of new external borrowing (in percent)	...	...	...	…	…	24.2	21.2	24.8	23.8	23.7	23.1	23.5	23.0	20.2	...

Sources: Country authorities; and staff estimates and projections.
1/ The consolidated public sector includes the central government, local governments, decentralized agencies, social security institutions, central bank, and public enterprises. Net debt is used.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2a.Honduras: Sensitivity Analysis for Key Indicators of Public Debt 2010-2030

	Projections							
	2010	2011	2012	2013	2014	2015	2020	2030
PV of Debt-to-GDP Ratio								
Baseline	22	24	25	25	25	25	23	21
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	22	23	24	24	25	25	23	22
A2. Primary balance is unchanged from 2010	22	25	28	30	32	34	42	57
A3. Permanently lower GDP growth 1/	22	24	25	26	27	28	32	55
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	22	25	28	30	31	32	36	46
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	22	25	28	29	29	29	26	25
B3. Combination of B1-B2 using one half standard deviation shocks	22	24	27	28	28	29	30	34
B4. One-time 30 percent real depreciation in 2011	22	30	31	31	31	31	29	30
B5. 10 percent of GDP increase in other debt-creating flows in 2011	22	34	35	35	35	35	32	29
PV of Debt-to-Revenue Ratio 2/								
Baseline	91	98	98	100	101	100	91	85
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	91	94	94	97	98	98	93	89
A2. Primary balance is unchanged from 2010	91	102	110	121	129	137	167	226
A3. Permanently lower GDP growth 1/	91	99	101	106	109	111	129	219
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	91	103	111	119	124	128	145	185
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	91	103	113	115	115	114	105	98
B3. Combination of B1-B2 using one half standard deviation shocks	91	100	106	111	113	115	119	135
B4. One-time 30 percent real depreciation in 2011	91	123	123	125	124	123	116	120
B5. 10 percent of GDP increase in other debt-creating flows in 2011	91	139	139	141	141	139	128	117
Debt Service-to-Revenue Ratio 2/								
Baseline	13	18	17	16	17	17	17	16
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	13	17	16	15	17	17	17	16
A2. Primary balance is unchanged from 2010	13	18	18	19	21	23	27	37
A3. Permanently lower GDP growth 1/	13	18	17	17	18	19	22	35
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	13	18	18	19	21	21	24	31
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	13	18	18	20	20	19	19	18
B3. Combination of B1-B2 using one half standard deviation shocks	13	18	17	18	19	20	20	23
B4. One-time 30 percent real depreciation in 2011	13	18	18	18	19	20	22	24
B5. 10 percent of GDP increase in other debt-creating flows in 2011	13	18	25	26	23	22	22	21

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table 3a.: External Debt Sustainability Framework, 2007-2030 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical 0 Average	Standard 0 Deviation	Projections									
	2007	2008	2009			2010	2011	2012	2013	2014	2015	2010-2015 Average	2020	2030	2016-2030 Average
External debt (nominal) 1/	**25.8**	**25.3**	**22.9**			**24.1**	**26.1**	**27.4**	**28.2**	**28.4**	**28.2**		**26.9**	**25.8**	
o/w public and publicly guaranteed (PPG)	16.4	17.1	17.2			19.2	21.3	22.5	23.1	23.1	22.8		20.5	17.7	
Change in external debt	-10.3	-0.5	-2.5			1.2	2.0	1.3	0.8	0.2	-0.2		-0.3	0.0	
Identified net debt-creating flows	-2.7	3.8	-1.5			1.2	1.3	0.9	0.5	0.3	0.1		-0.4	-1.7	
Non-interest current account deficit	**8.4**	**12.4**	**2.6**	**5.3**	**3.2**	**5.6**	**6.5**	**6.5**	**6.4**	**6.4**	**6.2**		**5.9**	**4.9**	5.7
Deficit in balance of goods and services	27.5	31.3	18.3			19.9	20.2	19.9	19.8	19.5	19.1		17.8	12.8	
Exports	53.2	52.7	42.2			42.7	43.0	44.0	45.3	46.0	46.3		48.8	57.8	
Imports	80.6	84.0	60.6			62.6	63.2	63.8	65.1	65.4	65.4		66.6	70.5	
Net current transfers (negative = inflow)	-21.5	-21.7	-19.3	-15.9	5.7	-17.2	-16.6	-16.2	-16.1	-15.8	-15.4		-14.3	-11.3	-13.3
o/w official	-1.2	-1.9	-1.4			-0.7	-0.7	-0.7	-0.7	-0.7	-0.7		-0.8	-0.4	
Other current account flows (negative = net inflow)	2.4	2.8	3.6			3.0	2.8	2.9	2.7	2.7	2.5		2.4	3.4	
Net FDI (negative = inflow)	**-7.5**	**-6.3**	**-4.2**	**-5.4**	**1.3**	**-4.5**	**-4.8**	**-5.2**	**-5.4**	**-5.6**	**-5.7**		**-5.9**	**-6.3**	-6.0
Endogenous debt dynamics 2/	**-3.6**	**-2.3**	**0.0**			**0.0**	**-0.3**	**-0.5**	**-0.5**	**-0.4**	**-0.4**		**-0.4**	**-0.3**	
Contribution from nominal interest rate	0.7	0.6	0.6			0.5	0.5	0.5	0.6	0.6	0.6		0.6	0.7	
Contribution from real GDP growth	-2.0	-0.9	0.5			-0.5	-0.8	-1.0	-1.1	-1.1	-1.1		-1.0	-1.0	
Contribution from price and exchange rate changes	-2.3	-2.0	-1.1			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**-7.6**	**-4.3**	**-1.0**			**0.1**	**0.7**	**0.3**	**0.3**	**-0.1**	**-0.3**		**0.1**	**1.8**	
o/w exceptional financing	-1.0	-1.2	1.9			-0.1	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	…	…	18.0			18.8	20.7	22.1	22.9	23.2	23.2		22.5	22.2	
In percent of exports	…	…	42.7			43.9	48.2	50.2	50.6	50.4	50.1		46.1	38.5	
PV of PPG external debt	**…**	**…**	**12.4**			**13.9**	**16.0**	**17.2**	**17.8**	**17.9**	**17.7**		**16.1**	**14.1**	
In percent of exports	**…**	**…**	**29.3**			**32.5**	**37.1**	**39.0**	**39.4**	**39.0**	**38.3**		**33.1**	**24.4**	
In percent of government revenues	**…**	**…**	**54.0**			**60.7**	**68.0**	**73.0**	**75.8**	**76.2**	**75.4**		**66.1**	**57.6**	
Debt service-to-exports ratio (in percent)	**4.1**	**5.0**	**9.1**			**5.2**	**4.7**	**4.6**	**4.5**	**4.7**	**4.9**		**4.7**	**3.0**	
PPG debt service-to-exports ratio (in percent)	**1.7**	**1.1**	**3.7**			**1.9**	**1.6**	**1.7**	**1.8**	**2.1**	**2.4**		**2.8**	**1.9**	
PPG debt service-to-revenue ratio (in percent)	**3.9**	**2.4**	**6.7**			**3.6**	**3.0**	**3.2**	**3.4**	**4.2**	**4.8**		**5.7**	**4.6**	
Total gross financing need (Billions of U.S. dollars)	0.7	1.6	0.8			0.7	0.8	0.8	0.8	0.8	0.9		1.1	1.3	
Non-interest current account deficit that stabilizes debt ratio	18.7	12.9	5.1			4.4	4.5	5.3	5.6	6.1	6.4		6.2	4.9	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.2	4.0	-1.9	3.8	2.9	2.4	3.5	4.0	4.0	4.0	4.0	3.7	4.0	4.0	4.0
GDP deflator in US dollar terms (change in percent)	6.9	8.5	4.5	6.5	9.5	4.9	2.7	0.9	0.2	1.5	2.3	2.1	2.1	2.1	2.1
Effective interest rate (percent) 5/	2.1	2.7	2.4	2.2	0.5	2.6	2.2	2.2	2.2	2.4	2.3	2.3	2.5	2.7	2.5
Growth of exports of G&S (US dollar terms, in percent)	9.0	11.7	-17.8	12.2	23.2	8.8	7.0	7.3	7.2	7.2	7.3	7.5	7.4	8.5	7.8
Growth of imports of G&S (US dollar terms, in percent)	19.4	17.5	-26.1	12.6	19.6	11.2	7.3	5.9	6.2	6.2	6.4	7.2	6.6	7.2	6.7
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	24.2	21.2	24.8	23.8	23.7	23.1	23.5	23.0	20.2	22.1
Government revenues (excluding grants, in percent of GDP)	22.8	24.2	23.0			22.9	23.5	23.5	23.5	23.5	23.5		24.4	24.4	24.3
Aid flows (in Billions of US dollars) 7/	0.5	0.7	0.5			0.6	0.4	0.5	0.5	0.5	0.5		0.4	0.8	
o/w Grants	0.2	0.3	0.3			0.3	0.2	0.3	0.3	0.3	0.3		0.2	0.3	
o/w Concessional loans	0.3	0.4	0.3			0.3	0.3	0.2	0.2	0.2	0.2		0.3	0.4	
Grant-equivalent financing (in percent of GDP) 8/	…	…	…			2.5	1.8	2.1	2.0	1.9	1.9		1.0	1.0	1.2
Grant-equivalent financing (in percent of external financing) 8/	…	…	…			48.8	38.2	53.1	57.3	58.0	58.7		43.2	43.9	46.4
Memorandum items:															
Nominal GDP (Billions of US dollars)	12.3	13.9	14.3			15.3	16.3	17.1	17.8	18.8	20.0		27.1	49.5	
Nominal dollar GDP growth	13.5	12.8	2.5			7.5	6.3	4.9	4.2	5.6	6.4	5.8	6.2	6.2	6.2
PV of PPG external debt (in Billions of US dollars)			1.8			2.1	2.6	2.9	3.1	3.3	3.5		4.3	6.7	
(PVt-PVt-1)/GDPt-1 (in percent)						2.5	2.8	1.8	1.4	1.1	0.9	1.8	0.6	0.8	0.7
Gross remittances (Billions of US dollars)	2.6	2.8	2.5			2.4	2.5	2.6	2.7	2.8	3.0		3.8	6.5	
PV of PPG external debt (in percent of GDP + remittances)	…	…	10.6			12.0	13.8	14.9	15.5	15.6	15.4		14.1	12.4	
PV of PPG external debt (in percent of exports + remittances)	…	…	20.8			23.7	27.3	28.9	29.4	29.3	29.0		25.7	19.8	
Debt service of PPG external debt (in percent of exports + remittances)	…	…	2.6			1.4	1.2	1.3	1.3	1.6	1.8		2.2	1.6	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 3b.Honduras: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030
(In percent)

	Projections							
	2010	2011	2012	2013	2014	2015	2020	2030
PV of debt-to GDP ratio								
Baseline	14	16	17	18	18	18	16	14
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	14	14	13	12	11	10	8	11
A2. New public sector loans on less favorable terms in 2010-2030 2	14	17	18	19	20	20	21	22
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	14	16	18	18	18	18	16	14
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	14	22	34	35	34	34	28	18
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	14	17	18	19	19	19	17	14
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	14	22	29	29	29	28	24	16
B5. Combination of B1-B4 using one-half standard deviation shocks	14	22	31	31	31	31	25	17
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	14	22	24	24	25	24	22	18
PV of debt-to-exports ratio								
Baseline	32	37	39	39	39	38	33	24
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	32	32	29	26	24	22	16	18
A2. New public sector loans on less favorable terms in 2010-2030 2	32	39	41	43	44	44	42	39
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	32	37	38	38	38	37	32	23
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	32	61	112	111	108	105	82	44
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	32	37	38	38	38	37	32	23
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	32	50	65	64	63	61	49	28
B5. Combination of B1-B4 using one-half standard deviation shocks	32	52	77	76	75	73	57	32
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	32	37	38	38	38	37	32	23
PV of debt-to-revenue ratio								
Baseline	61	68	73	76	76	75	66	58
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	61	59	55	51	47	43	33	43
A2. New public sector loans on less favorable terms in 2010-2030 2	61	71	77	82	85	86	85	92
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	61	69	75	78	78	77	67	57
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	61	93	145	147	146	143	113	72
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	61	71	78	81	81	80	69	59
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	61	92	121	124	123	121	97	66
B5. Combination of B1-B4 using one-half standard deviation shocks	61	92	130	133	132	130	104	69
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	61	95	100	104	105	103	89	76

Table 3b.Honduras: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030 (continued)
(In percent)

Debt service-to-exports ratio

Baseline	2	2	2	2	2	2	**3**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	2	2	1	1	2	2	**1**	1
A2. New public sector loans on less favorable terms in 2010-2030 2	2	2	2	2	2	2	**3**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	2	2	2	2	2	2	**3**	2
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	2	2	3	4	5	5	**7**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	2	2	2	2	2	2	**3**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	2	2	2	3	3	3	**4**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	2	3	3	4	**5**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	2	2	2	2	2	2	**3**	2

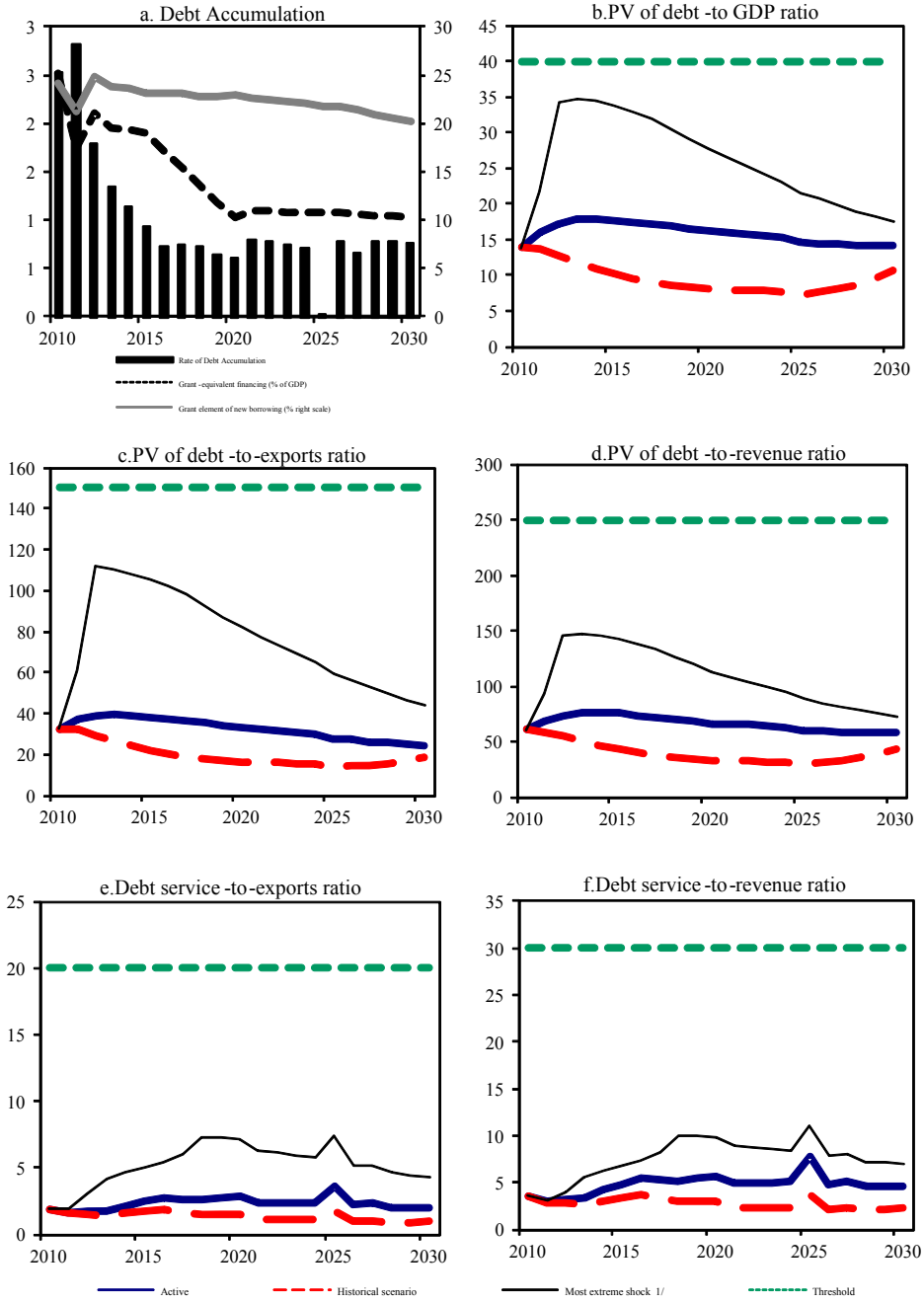
Debt service-to-revenue ratio

Baseline	4	3	3	3	4	5	**6**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	4	3	3	3	3	3	**3**	2
A2. New public sector loans on less favorable terms in 2010-2030 2	4	3	3	4	4	4	**6**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	4	3	3	4	4	5	**6**	5
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	4	3	4	6	6	7	**10**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	4	3	4	4	5	5	**6**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	4	3	4	5	6	6	**9**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	4	3	4	5	6	6	**9**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	4	4	5	5	6	7	**8**	6
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	16	16	16	16	16	16	**16**	16

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

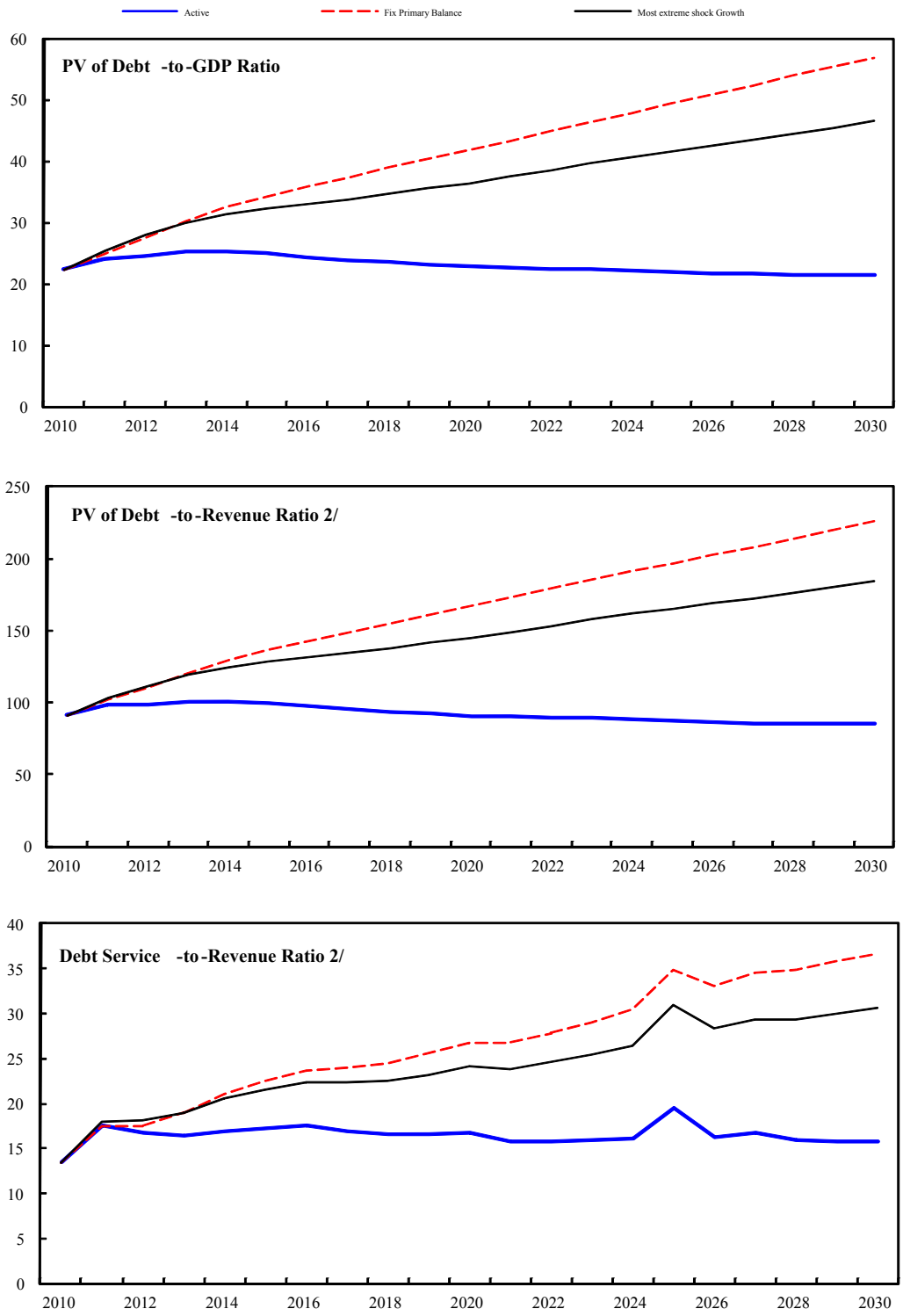
Figure 1a. Honduras: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds a Exports shock; in c. to a Exports shock; in d. to a Exports shock; in e. to a Exports shock and in figure f. to a Exports shock

Figure 2a.Honduras: Indicators of Public Debt Under Alternative Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020.

2/ Revenues are defined inclusive of grants.